

ESE 101:

Homework 4 (due November 6, 2025):

1 Pasadena Air Pollution

Air pollution is a serious problem in Los Angeles. In addition to well known health problems, like increased cancer and asthma rates, the smog can also darken the skies by scattering incoming shortwave radiation. Not only is `aq.caltech.edu` a good site to check the current air quality, but it also lists live concentrations of two common shortwave scatters, PM2.5 and PM10 taken on top of Caltech tower. PM2.5 indicates the concentration of particulate matter below 2.5 microns (10^{-6}m) while PM10 indicates the concentration below 10 microns (which includes PM2.5). In this question we will compute the optical depth and estimate the affect of these two particles on radiation reaching the surface. We'll assume a gray atmosphere and ignore effects on shortwave from other absorbers and scatterers.

1. Visit `aq.caltech.edu` and tell us the time you checked and what the concentrations of PM2.5 and PM10 are. Compute the difference to estimate the fraction of absorbers greater than 2.5 microns. Please convert to SI units.
2. Assume the mass extinction coefficients, $k_e^{\text{PM2.5}} = 4\text{m}^2\text{g}^{-1}$ and $k_e^{\text{PM2.5-10}} = 1\text{m}^2\text{g}^{-1}$, compute the volume extinction coefficients, $\alpha_e^{\text{PM2.5}}$ and $\alpha_e^{\text{PM2.5-10}}$, and total volume extinction coefficient, α_e , in SI units at Caltech tower. Explain what scattering regime these particles are in and why the extinction coefficient for the smaller particles is larger.
3. Assume the planetary boundary layer over LA (the height to which the particulates are mixed) is 1km. Compute the vertical optical depth by integration. Use a scale height of 7km.
4. Compute the fractional and absolute reduction in solar radiation from TOA to the surface at solar noon on the day you do this assignment.

2 Heating Rate for Well-Mixed Absorber

Suppose a gas that absorbs solar radiation has a uniform mixing ratio of 1 g kg^{-1} and an absorption cross section of $5 \text{ m}^2 \text{ kg}^{-1}$. Assume an isothermal atmosphere with $T = 260 \text{ K}$, and a surface pressure of 10^5 Pa .¹

1. At what altitude is the rate of energy absorption per unit volume maximal when the sun is directly overhead?
2. At what altitude is the rate of energy absorption per unit volume maximal when the solar zenith angle is 45° ? What difference does the zenith angle make?
3. If the frequency range over which absorption takes place contains 5% of the total solar energy flux, what is the heating rate (K day^{-1}) at the level of maximum energy absorption? What is the heating rate one density scale height above and below the level of maximum absorption? (Use globally averaged insolation.)
4. Give two examples of well-mixed absorbers and two examples of absorbers that are not well mixed in earth's atmosphere.

¹From Hartmann, *Global Physical Climatology*.